Mahmoud Salakawy

AI & ML Engineer

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PROFILE

I build ML and GenAI systems that detect network anomalies, ground evidence, and support engineering decisions.

10 years in telecom RAN. 4+ years building machine learning systems.

Time-Series ML / GenAI / RAG / Telecom AI

EXPERIENCE

AI & automation | 2024 - PRESENT

Part-time volunteer contribution alongside a core RAN role: ML pipelines, grounded retrieval, text classification and mentoring.

Radio expert · RAN integration | 2018 - PRESENT

Telecom domain experience informing anomaly analysis, explainability and repeatable data workflows.

RAN integration engineering | 2017 - 2018

A foundation in radio network integration, troubleshooting and engineering review.

SELECTED ENGINEERING WORK

IT Ticket Classification

From unstructured support text to consistent, actionable categories. A text-classification pipeline with a typed prediction API.

Tools: scikit-learn, FastAPI, Pydantic

5G QoS Classification

Predicting application type through telecom feature engineering. A comparison of tree-based models with telecom-informed features.

Tools: XGBoost, LightGBM, SHAP

Resume Screening Automation

Structured information extraction and fit scoring in a reviewable workflow. An extraction and assessment workflow with structured outputs.

Tools: n8n, LLMs, Structured outputs

EDUCATION AND CERTIFICATION

BSc Electrical Engineering, Cairo University, 2011. SAFe 6.0 Practitioner.

RAN Anomaly RCA

Detection research and a separate grounded RCA prototype. This case study does not represent a single system with all capabilities deployed together.

ANOMALY DETECTION

- Per-cell baselines with Prophet and SARIMAX; complementary Isolation Forest detection.
- Severity ranking and persistence checks distinguish sustained degradation from isolated events.
- SHAP explanations and three report views: executive summary, site overview and cell detail.
- Excel / CSV inputs; offline analysis without live network access.

GROUNDED ROOT-CAUSE ANALYSIS

- Five-step retrieval flow: understand query, look up anomaly, form search, retrieve and re-rank, respond.
- FAISS retrieval with KPI- and technology-aware re-ranking.
- Grounding gate: no anomaly evidence means no RCA verdict. Facts and hypotheses stay separate.
- Deterministic mock mode and optional provider-agnostic LLM mode; safety-contract tests for the prototype.

RESEARCH

Policy-Calibrated Forecast Intervals for Anomaly Detection in Radio Access Mobile Network

M. A. ElSalakawy, A. H. N. Ahmed, M. R. Abonazel

Journal of Intelligent Computing and Data Science | JICDS · Vol. 2 (1), pp. 48-73 · 2026

Policy-calibrated forecast intervals connect anomaly alerts to engineering relevance.

Published evaluation: 91.27%. Balanced accuracy on the paper's expert-labelled evaluation set (Table 11). A research result, not a guarantee for other datasets or the RCA prototype.

Published research / DOI